FOURIER-MUKAI FUNCTOR PRESERVES BOUNDED COHERENT IN FOUR CASES

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Pseudo-coherence chain. cohomology-lemma-pseudo-coherent-pullback, perfect-lemma-identify-pseudo-coherent-noetherian, cohomology-lemma-tor-pseudo-coherent (cross-chapter, named). Shows the twisted pullback is pseudo-coherent with coherent cohomology.
- Direct image coherence. perfect-lemma-direct-image-coherent (cross-chapter, named). Reduces boundedness of the FM image to boundedness of the tensor product.
- Tor amplitude tools. cohomology-lemma-tor-amplitude-stalk, more-algebra-lemma-complex-finite-tor-dimension-modules (cross-chapter, named). Cases (3) and (4) verify finite tor dimension stalkwise.

Source reference. Original reference 00OE: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09U7: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09J3: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09U9: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DE: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original reference 066H: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E2: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E8: focused-seeds/geometry/S9-eqv-fourier-mukai-Coh/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a Noetherian scheme. Let X and Y be schemes of finite type over S. Let $K \in D^b_{Coh}(\mathcal{O}_{X\times_S Y})$. The corresponding Fourier-Mukai functor Φ_K sends $D^b_{Coh}(\mathcal{O}_X)$ into $D^b_{Coh}(\mathcal{O}_Y)$ if at least one of the following conditions is satisfied:*

1. *the support of $H^i(K)$ is proper over Y for all i, and K has finite tor dimension as an object of $D(pr_1^{-1}\mathcal{O}_X)$,*
2. *K can be represented by a bounded complex $\mathcal{K}^\bullet$ of coherent $\mathcal{O}_{X\times_S Y}$-modules, flat over X, with support proper over Y,*
3. *the support of $H^i(K)$ is proper over Y for all i and X is a regular scheme,*
4. *K is perfect, the support of $H^i(K)$ is proper over Y for all i, and $Y \to S$ is flat.*

Furthermore in each case the support condition is automatic if $X \to S$ is proper.

Proof. Let M be an object of $D^b_{Coh}(\mathcal{O}_X)$. In each case we will use Derived Categories of Schemes, Lemma 08E2 to show that

$$\Phi_K(M) = R\mathrm{pr}_{2,*}(L\mathrm{pr}_1^* M \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K)$$

is in $D^b_{Coh}(\mathcal{O}_Y)$. The derived tensor product $L\mathrm{pr}_1^* M \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K$ is a pseudo-coherent object of $D(\mathcal{O}_{X\times_S Y})$ (by Cohomology, Lemma 09U7, Derived Categories of Schemes, Lemma 08E8, and Cohomology, Lemma 09J3) whence has coherent cohomology sheaves (by Derived Categories of Schemes, Lemma 08E8 again). In each case the supports of the cohomology sheaves $H^i(L\mathrm{pr}_1^* M \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K)$ is proper over Y as these supports are contained in the union of the supports of the $H^i(K)$. Hence in each case it suffices to prove that this tensor product is bounded below.

Case (1). By Cohomology, Lemma 08DE we have

$$L\mathrm{pr}_1^* M \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} K \cong \mathrm{pr}_1^{-1} M \otimes^{\mathbf{L}}_{\mathrm{pr}_1^{-1}\mathcal{O}_X} K$$

with obvious notation. Hence the assumption on tor dimension and the fact that M has only a finite number of nonzero cohomology sheaves, implies the bound we want.

Case (2) follows because here the assumption implies that K has finite tor dimension as an object of $D(\mathrm{pr}_1^{-1}\mathcal{O}_X)$ hence the argument in the previous paragraph applies.

In Case (3) it is also the case that K has finite tor dimension as an object of $D(\mathrm{pr}_1^{-1}\mathcal{O}_X)$. Namely, choose affine opens $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ of X and Y mapping into the affine open $W = \mathrm{Spec}(R)$ of S. Then $K|_{U\times V}$ is given by a bounded complex of finite $A \otimes_R B$-modules $M^\bullet$. Since A is a regular ring of finite dimension we see that each M^i has finite projective dimension as an A-module (Algebra, Lemma 00OE) and hence finite tor dimension as an A-module. Thus $M^\bullet$ has finite tor dimension as a complex of A-modules (More on Algebra, Lemma 066H). Since $X \times Y$ is quasi-compact we conclude there exist $[a, b]$ such that for every point $z \in X \times Y$ the stalk K_z has tor amplitude in $[a, b]$ over $\mathcal{O}_{X,\mathrm{pr}_1(z)}$. This implies K has bounded tor dimension as an object of $D(\mathrm{pr}_1^{-1}\mathcal{O}_X)$, see Cohomology, Lemma 09U9. We conclude as in the previous to paragraphs.

Case (4). With notation as above, the ring map $R \to B$ is flat. Hence the ring map $A \to A \otimes_R B$ is flat. Hence any projective $A \otimes_R B$-module is A-flat. Thus any perfect complex of $A \otimes_R B$-modules has finite tor dimension as a complex of A-modules and we conclude as before. □